

GLaDOS

Geometric Lattice Deduction Over Streams

*a sound, abstain-capable deduction organ — and the evidence that an LLM can be made to **causally** reason through it*

ember + Claude (Opus 4.8) · draft 2026-06-26

Honesty banner. We write “empirically sound on this distribution,” never “certified.” Soundness here is a **property of an output check we run**, not a theorem about the network. The borrowed parts are named as borrowed: the lattice / abstract-interpretation narrowing + abstain + on-policy targets are the Lattice Deduction Transformer (LDT, arXiv:2605.08605); the coupling primitives (zero-init gate, structured readback, freeze-the-reasoner) are TransNAR (arXiv:2406.09308); the process-reward recipes are RLTT (arXiv:2602.10520) and LSRL (Findings EMNLP 2025). What is **ours** is the combination and the discipline: a per-problem-compiled, output-checked deductor woven into a pretrained LLM’s residual stream and verifier-trained, with adversarial causal controls run on every result. Negatives in this paper are results, not omissions.

Abstract. We ask whether a learned, differentiable, output-checked deduction module can be woven into a pretrained language model so that the model’s own generation reasons **through it** — and we hold ourselves to the standard that broke the closest prior work (SATNet): a sound module can still be **bypassed**, so only causal controls (shuffle / permute / corrupt the module’s state) prove genuine deduction. We build an exact CSP harness as ground truth, a **bank** of deduction organs wired to one typed spine (certified arc / factor / modular / GF(2) / macro reductions + a verifier-gated neural narrower + forward-chaining + energy), and a staged training recipe. Our findings, with the negatives: coupling a frozen LLM to a checked deductor gains calibrated abstention it lacks (88.8% vs 1.4%); RLVR fixes **calibration**, not **capacity**; 7B behaves like 1B (capacity is not the wall); **cold co-training fails** — but a staged recipe (organ → excellence → freeze → weave) makes a **learned, general** organ causally load-bearing. The consolidated live organ — latent $\alpha \rightarrow$ a verifier-gated reduced product over a **certified floor** → rich-state γ — **engages** (corrupt → drop 97 pts, lift +77), is **miscompile-robust by construction** (the floor cannot be poisoned), **generalizes by reduction** (a no-direct-faculty family solved 100% via a Karp edge), and is **legible learned-free** (α -compile read at F1 0.90 / recall 0.98); the graft is **residual-stream-agnostic** (7/7 bases incl. a Mamba-hybrid). We state the honest scope plainly: in the easy regime the certified floor carries correctness on its own, so the bottleneck test is saturated — the open frontier is α compiling from **real natural language** with no provided structure. We position this against TransNAR, Coconut, SATNet, and the process-reward line.

1 The bet, in one paragraph

Language models confabulate because nothing in their forward pass is allowed to say “I cannot conclude that.” The bet of GLaDOS is that you can give a transformer a **reasoning channel that can abstain** by welding a small, learned deduction module into its residual stream — and that the module can stay **reliable** not because it is a verifier, but because we **check its output**. The check is the whole trick. A learned module is free to be loose, general, even emergent; if a cheap exact check sits at the boundary, the only outputs that survive are the correct ones, and the rest become honest abstentions. We call the module an *organ* and the assembled system GLaDOS.

The reason to want this **woven**, rather than as an external SAT/SMT tool the LLM calls in text, is bandwidth. A text round-trip to a solver is non-differentiable and lossy: the LLM never sees the solver’s internal certified state, gets no gradient on “did I state the problem right,” and inherits none of the solver’s $\perp$ (don’t-know) channel. Weaving keeps the interface differentiable end-to-end and lets the LLM’s **own** head generate from a hidden state already saturated by the deduction.

1.1 The spine

The organ’s state a_t is a point in a per-position *powerset lattice* — for each variable, the set of candidate values still alive. One step of deduction proposes, then commits only by **narrowing**:

$$a_{t+1} = a_t \cap \Pi(f_\theta(a_t, p, m))$$

Here f_θ is a **learned** cell (it proposes, loosely), Π projects the proposal into the lattice, and the meet $\cap$ guarantees $a_{t+1} \subseteq a_t$: the step can only **remove** candidates, never add one. The bilinear/learned part **proposes**; the lattice meet is the sole **authority** on elimination. That single algebraic fact — monotone narrowing — is what makes the recurrence sound-by-construction **early in training** (it is the training-wheels inductive bias), and it is what an exact verifier at the output can later relax: the endpoint is a general, emergent reasoner that stays sound because the **boundary** checks it, not because every internal step is provably correct.

1.2 The one-sentence architecture

At layer k of the host LLM, a learned map α **compiles** the hidden state into a *typed factor-graph program* — variables, domains, and factors, **not** a pre-solved answer; the **organ narrows** that program recurrently, **between** host layers (no backprop through repeated host passes); a learned map γ **writes** the certified state **densely and structurally** back into the residual stream through a **zero-init gate**; the host’s **LM head generates** the answer from the saturated hidden state; an **exact verifier** supplies the reward.

Two phrasings of the same discipline run through the whole paper. First: **soundness is a permission, not a prescription**. The output check buys correctness-given-the-organ; it lets us make the organ loose, learned, and general instead of a hand-built verifier. Second — and this is the lesson that cost us the most — **a sound organ can still be bypassed**. The check guarantees the answer is right **if the organ produced it**; it says nothing about whether the organ, versus the prompt text or a LoRA, actually produced it. Only adversarial causal controls (shuffle / permute / corrupt the organ’s state and watch accuracy collapse) prove the deduction is load-bearing. We run them on every result, and they are the real contribution.

2 The harness: an exact world to be sound about

Before any neural claim, we need ground truth — a generator of constraint problems where the **exactly correct** deduction is computable, so “the organ eliminated a true candidate” is a measurable event, not a matter of opinion. The harness (`clair/csp.py`, `clair/levels.py`) is that world, and it is deliberately unglamorous: its job is to make our soundness claims falsifiable.

2.1 Exact deduction, by construction

For a constraint satisfaction problem p , the harness computes the **exact deduction closure** ded_p : the per-variable set of values that appear in **at least one** full solution. This is the tightest sound narrowing possible — anything tighter would kill a value that some solution needs. Two facts about it are load-bearing and are grounded in the LDT line’s Lean formalization (arXiv:2605.08605):

- **Soundness is free if you check; completeness is the hard part.** A narrowing that only removes values no solution uses is sound by definition. The interesting, **learnable** quantity is *completeness*: how close to ded_p the organ gets — i.e. whether it reaches a checkable answer or honestly abstains. Completeness, not soundness, is the research variable.
- **Per-cell arc-consistency equals exact deduction only at the right level.** Cheap local propagation (per-cell AC) recovers ded_p on binary-structured problems but **not** on problems with higher-arity structure. The harness exposes this as a lattice of *abstraction levels*: level-0 is per-cell/unary, level-1 is pairwise, level-2 carries genuine factor (3-way+) structure.

2.2 The general factor lattice, and the affine wall

The original LDT formulation is binary (pairwise exclusion). Our harness generalizes it to a *factor lattice of any arity*: a factor is a tuple of variables plus an allowed-tuples table, and narrowing propagates through factors, not just edges. This matters because of one recurring obstacle we name the *affine wall*.

The affine wall = width. Three-way / modular-arithmetic / XOR-like constraints (the “affine” structure, e.g. $x + y + z \equiv 0$) cannot be captured by **any** amount of pairwise propagation: per-cell AC is provably stuck. They require a **grade-3 / ternary** primitive in the per-step computation. In the harness, the factor lattice “breaks XOR’s affine wall at level-2” — the factor structure is exactly what supplies the missing arity. The wall is therefore a **width** (per-step expressivity) limit, not a **depth** one; see §4 and the macro-deduction result for why depth alone cannot manufacture it.

2.3 The polymorphism diagnostic

Constraint tractability has a classical algebraic signature: a CSP is tractable iff its constraint language admits certain **polymorphisms** (closure operations). The harness ships a polymorphism diagnostic so that, for any generated instance class, we can tell **in advance** whether per-cell propagation should suffice or whether factor-level structure is required. This is what lets us aim experiments at the regime a method targets — a discipline lesson learned the hard way (don’t conclude a method failed from a mis-aimed run).

2.4 The verifier ladder

On top of the CSP core sits a ladder of exactly-verifiable tasks that double as the training curriculum and the generality test (`curriculum.py`, `fol.py`, `smt.py`, `schedule.py`): a forward-chainer giving entail / contradict / unknown labels, a z3 SMT oracle, a multi-task scheduler, and Bedrock-rendered natural-language surface forms so the same logical problem appears in many phrasings. The diversity **is** the generality measurement: hold out a phrasing or a problem size, and OOD soundness becomes observable.

3 The woven organ

The architecture is one forward pass up a **frozen** host’s layer stack, with the organ spliced in as a recurrent side-loop at the residual stream. The temptation — and the trap — is to make α powerful enough to **solve** the problem and γ wide enough to **dump** an answer. A GPT-5.5 review of an early “dense weave” caught exactly this: unstructured dense surgery into the residual stream produces a **new** bypass, where the LM encodes the answer through a lattice-shaped tensor without ever learning stable deduction. The corrections below are not cosmetic; each closes a bypass. The coupling is **residual-stream-agnostic** — the same math grafts onto every base we tried (§“the graft”).

3.1 α compiles latently — the verifier, not α , guards soundness

α (`latent_organ.DenseLatentProjector`) maps the host’s hidden state to the organ’s input: a **per-cell candidate lattice** (`b0` logits over the cells’ value sets). It is a **dense, fully-latent compile** — no symbolic extraction, no parse of the prompt into a hand-specified program. This is the part with **no** prior art — TransNAR’s reasoner ingests a clean external graph the LLM never has to produce (§5); α is the unprecedented half, and the staged recipe gives it its own direct objective.

The early design forbade α from emitting survival logits, fearing “the answer posterior shaped like a lattice” as a bypass. The consolidated organ removes that fear **structurally** rather than by prohibition: α ’s compile is presented to the composer as **one verifier-gated proposal among many**, sitting on a **certified floor**. A confidently-wrong α can only ever sharpen the composed lattice **toward** the exact per-cell transformer, never drop a value a real solution uses — so it **cannot**

poison the state, and there is no answer-shaped bypass to camp in. The constraint **structure** the bank consumes comes from the problem instance (threaded into the records), not from α ; α supplies only the latent per-cell lattice.

3.2 The bank + composer narrows between layers — the host is not the recurrence

Between two host layers, the organ slot runs the **bank as a verifier-gated reduced product** (the live artifact is `organ.bank_woven`). On the problem’s true structure it composes:

- the **certified floor** — sound-by-construction CSP-domain reductions: arc-consistency, level-2 factor consistency, modular Smith-normal-form, GF(2) row-space, reach-doubling macros (`organ.bank`);
- the **pretrained** neural narrower (`core_narrow_organ`, the union-trained general deductor) and α ’s own compile — both **verifier-gated** by the exact per-cell transformer (`csp.exact_dedP`), so a wrong neural proposal is intersected back to what the verifier confirms.

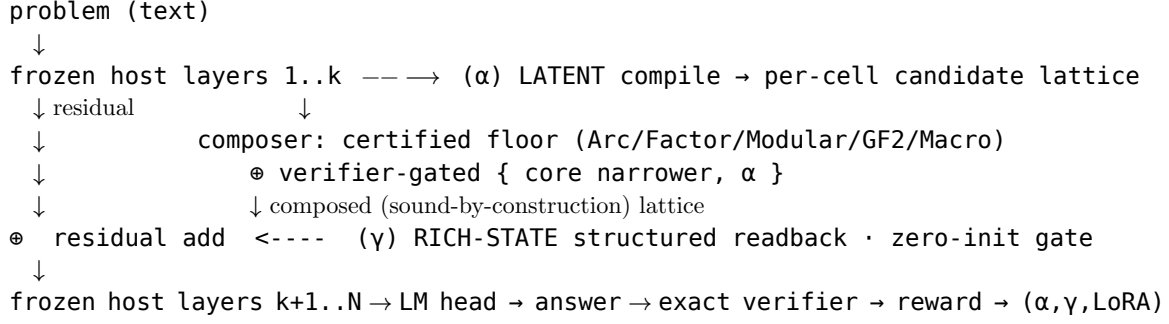
The review’s correction made this non-optional: output-verification makes accepted **answers** sound, not intermediate **states** — “one unsound reduction poisons the next organ before the final check.” So composition is a **reduced product** (`compose.reduced_product`), each reducer with its own soundness rule, and the **meet of sound narrowings is sound by construction**. The result: the injected lattice is **miscompile-robust by construction**, which **subsumes** the earlier “train the organ to be robust” calibration story.

We **do not** backprop through repeated full host passes; the composer is discrete and its output is detached (γ + LoRA learn to read it, while α is grounded by its own direct loss). The host drifts the state coarsely across blocks; the organ refines it finely within the loop — a two-scale split matching the dynamics of recurrent-depth transformers (arXiv:2509.23314). Halting is **monotone**: candidate-set cardinality only decreases, so a cardinality-plateau is a clean early-exit.

3.3 γ reads back richly and structurally, through a zero-init gate

γ writes the **composed** lattice state back into the residual stream. Three design choices, the first two corroborated by TransNAR (arXiv:2406.09308):

- **Structured, not flattened.** γ uses equivariant adapters over the explicit cell-candidate tensors, not a size-tied flatten (which is anti-equivariant and wrecks size generalization).
- **Zero-init gate.** The readback is gated **closed** (Flamingo / LLaMA-Adapter style), so step-0 of the hybrid is the host **exactly** — a **bitwise no-op at init** — and the model learns to open the gate only as the organ proves useful.
- **Rich-state.** γ reads the full candidate **set** per cell **plus** $\llbracket \text{set} / d, \text{reliability} \rrbracket$ — the alien-consumer finding: the LM mines a partial-but-sound lattice and discounts low-reliability cells, so an α -miscompile degrades **gracefully** instead of dragging the LM off a cliff (rich beats a collapsed point-readout by +27.5 pts on partial states).



Listing 1: The weave. α compiles latently (it does not solve, and cannot poison); the bank+composer narrows between layers on a certified floor (the host is not the recurrence); γ reads the **composed** lattice back richly through a closed-init gate; the host’s own LM head generates; an exact verifier closes the loop. Only α , γ , and the host LoRA train — the base and the organ bank are frozen.

3.4 The organ bank: a shared protocol, many inference shapes

There is not one organ but a **bank**, every beast wired to one typed spine (organ.protocol.Reduction): an abstract STATE, a sound **reduce** (or a no-op = ABSTAIN), a **certificate** (sound-by-construction / neural-guidance / approximate), and the uniform `survival(state, K)` readout γ projects. The status partition is honest:

- **certified, CSP-domain** (the reduced-product floor): arc / factor consistency, modular SNF, GF(2) row-space, reach-doubling macros, and the exact per-cell transformer as the **verifier**;
- **certified, own state type**: forward-chaining least-Herbrand entailment (`unification_chain`, the narrowing dual — grows facts);
- **approximate, own state type**: mean-field energy / equilibrium optimisation (the optimisation shape the narrowers cannot express);
- **neural-guidance, verifier-gated**: the union-trained `core_narrow_organ` and the blade/grade-prior affine deductor.

No single beast dominates; the verifier — never the LM router alone — arbitrates which narrowing is trusted.

3.5 Reductions: a cross-type faculty (one faculty + edges = many problems)

The composer routes **within** a state type. A second, distinct typed object — `organ.reductions.ProblemReduction` — routes **across** types: it maps an X-instance to a Y-instance of a different faculty and decodes the Y-solution back. The shipped Karp neighbourhood ($\text{MIS} \leftrightarrow \text{VC} \leftrightarrow \text{clique}$; $3\text{-SAT} / 2\text{-SAT} / \text{XOR-SAT} \rightarrow \text{CSP}$; $3\text{-SAT} \rightarrow \text{MIS} \rightarrow \text{Ising}$; $\{\text{MIS}, \text{MaxCut}, \text{partition}, \text{coloring}\} \rightarrow \text{Ising}$) reuses the exact Lucas $\text{NP} \rightarrow \text{Ising}$ encoders already in the repo; **every edge is exact-verified end-to-end against X’s own solver**, and `ReductionGraph.route` is Dijkstra cost-routing with a certified-over-approximate tie-break. The model learns **recognize-X · reduce-X→Y · solve-via-Y · decode** — the structural Karp prior the GNN-for-NP papers ignore — so one faculty plus edges covers everything reducible to it.

3.6 Geometric algebra, placed precisely

Geometric algebra was the project’s orphaned opening thread, and as a **generic channel mixer** it is a clean null (it ties SwiGLU/MLP in language and vision). Its real home is the factor deductor, via a structural identity: **geometric-product grades correspond to constraint arities**. Grade-0 (inner) $\leftrightarrow$ unary; grade-2 (the wedge $u \wedge v$, “these two cannot coexist”) $\leftrightarrow$ binary exclusion; grade-3 (the trivector $u \wedge v \wedge w$) $\leftrightarrow$ the ternary/affine primitive that breaks the level-0 wall. The right abstraction is the **un-truncated fold** — one wedge spanning all arities up to a capacity K .

Crucially, GA is a **completeness / representation** bias, **not** a soundness fix: soundness comes only from the output check and the certified kernels, never from the algebra (arXiv:2601.06793).

3.7 The staged recipe — the central lesson

Stage 1 — the organ to excellence, separately. Train the deductor hard across many tasks on the soundness-asymmetric dominate- ded_p loss until it is a general, sound, near-oracle **standalone** narrower (no LM). Cheap: pure organ. The recipe (from the sweep): 1.5M params, $R = 12$, 30% composed.

Stage 2 — freeze it, then weave. Graft the frozen bank+composer organ into a host; train LoRA + latent- α

1. zero-init γ on the answer-span LM cross-entropy, with the **engagement mechanism**: the two-stream lever (the LM generates from a fact-ablated prompt so the organ is the only route) + J_0 direct α -supervision (the SATNet grounding fix). Each translator competent **before** it must cooperate.

Stage 3 — RLVR, then consolidate. RL-from-verifiable-rewards (the organ is its own exact process reward), then re-run the host’s own recipe (Dolma $\oplus$ reasoning) **with the organ in place** so organ-use becomes native, not bolted-on.

Listing 2: The staged recipe. The central finding (§4) is that you must **not** cold-co-train the LM and the organ — two alien substrates, both bad at init, settle into a text-shortcut. Decouple: make the organ **good** (pretrained, frozen) and **necessary** (the easy path removed). The organ becomes load-bearing only when both hold.

3.8 The organ as its own process reward

Because the organ narrows **monotonically** and **checks** each step, it is its own exact per-step grader: the process-reward “progress” signal is the candidate-set cardinality drop at step t , and the “quality” signal is the soundness of that drop (guaranteed by the check). This is exactly the dense per-step supervision the process-reward line (RLTT/LSRL, §5/§6) had to approximate with a hackable LLM judge — here it is free, exact, and unhackable (organ.process_reward, mixed $0.7 \cdot \text{outcome} + 0.3 \cdot \text{process}$).

4 Training: the staged, oracle-supervised recipe

The architecture (§3) says **what** the weave is; this section says **how** it is taught. The recipe is **staged, curriculum-driven, and oracle-supervised** — and that last word matters: it is **not** classic semisupervised learning. Every label is **free and exact**. Witness-first generation hands us a known solution, and the exact per-cell deductor (clair.csp.exact_dedP) hands us the exact narrowing targets, the answers, and the verifiable rewards. There are **no human labels and no noisy pseudo-labels** — the “curriculum” is purely the **staging order**, the readout-forcing schedule, and the domain / composition / N mix. The entry point is `clair/organ/train.py` (organ for Stage 1, **weave** for Stages 2–3).

4.1 Stage 1 — pretrain the organ, standalone

The deductor learns to **narrow soundly**, with the base LLM not yet involved. The loss is **dominate-ded_p**: a soundness-asymmetric supervised loss against the exact per-cell transition. Eliminating a value that some solution actually uses (an **unsound** drop) is penalized heavily ($w_{\text{pos}} \approx 6$); keeping an extra candidate (**loose but sound**) is penalized lightly ($w_{\text{neg}} \approx 0.5$). The step is monotone — survival only ever decreases (the meet) — so the asymmetry trains the organ toward the sound side of the boundary. Training is **on-policy**: each batch is the previous step’s narrowed lattice rolled forward by the organ, with targets re-computed by the exact deductor every pass (the **given** set is recomputed after each meet).

The data is the **broad multi-domain corpus**: all beast generators (CSP families $\cdot$ xor/GF2 $\cdot$ modular $\cdot$ graph $\cdot$ permutation $\cdot$ type $\cdot$ FOL $\cdot$ optimize), plus **union relations** (named families

and random allowed-tuple tables, added **additively** — this is the fix for OOD-novel-relation soundness), wide N , and **30% composed problems** (multi-skill). The sweep recipe is **1.5M parameters**, **$R = 12$ recurrent steps** (with an $R = 8$ floor) and the 30% composition mixup. Composition **recall** is nearly free (the meet composes); composition **coverage** buys **soundness** on novel compositions — a need that **grows with scale**. The output is a frozen, general, sound narrower checkpoint (`runs/general_organ_full.pt`).

4.2 Stage 2 — freeze the organ, weave the live readout

Now the LLM learns to **generate through** the frozen organ. The Stage-1 organ is **frozen** (a slow-thaw / EMA variant is studied; default is freeze), the OLMo base is **frozen**, and the only trained parameters are **LoRA + latent- α + γ** :

- **latent- α** is a dense projection from the OLMo hidden state to the organ’s input tensors. It is **fully latent** — there is **no symbolic extraction step**. The hidden state compiles directly to the typed factor-graph program (variables, domains, factors), and never to the answer.
- γ is the structured readback, with its gate **zero-initialized** (closed) so step-0 of the hybrid is the host exactly.

The forward pass is: OLMo hidden $\rightarrow$ latent- α $\rightarrow$ the frozen organ narrows (under `no_grad`, so we **never** backprop through repeated OLMo passes) $\rightarrow \gamma$ injects the narrowed lattice into the residual $\rightarrow$ the LM head generates the answer. The loss is plain **LM cross-entropy on the answer span only**. α never gets a direct target; it learns purely from the gradient flowing **through the organ’s downstream success** — the LM does better when α has compiled a usable problem, and that signal is what teaches the compile.

The curriculum here is **readout-forcing**. Training starts in **cells-mode** — no facts in the prompt, so the organ is the **only** route to the answer — which forces the γ -decode to form **before** the LM can settle into a text-shortcut basin. Full-text problems are then mixed back in. The output is a woven model in which the LM **causally** wields the organ: corrupt the lattice and generation collapses.

4.3 Stage 3 — RLVR and consolidation

Two moves sharpen and stabilize the woven model:

- **RLVR** (GRPO; verifiable reward = the exact checker) with the **organ-as-process-reward** (LSRL-style: per-step candidate-set cardinality drop), mixed **0.7 outcome / 0.3 process** so the per-step signal shapes rather than dominates. This sharpens **calibration**, not capacity (§4); we report pass@1 **and** pass@k with ProRL and random-reward controls, on OLMo as the honest base (§6).
- **Consolidation**: continued-pretrain on **Dolma $\oplus$ a reasoning corpus** (general $\oplus$ organ-shaped, tuned ratio) with the organ in place, so organ-use folds into general behavior **without** catastrophic forgetting — the preserve-and-generalize endpoint.

4.4 The honest correction: readout-isolation vs live latent- α

Scope correction. The woven numbers reported in §4 — woven 98.8%, corrupt $\rightarrow$ 2.7% with full text present — are **readout-isolation** results: the organ is handed the **true** narrowed lattice (the ground-truth CSP, compiled outside the model), and what is proven is that the LM **reads back and generates through** a certified state it cannot get from the prompt. That readout is real and causally controlled, and it is the hard half TransNAR also enjoys (a clean external graph it never has to produce). The **real** woven model closes the remaining gap with **live latent- α** : α compiles the organ’s input **from the OLMo hidden state**, with **no symbolic extraction and no ground-truth CSP**. That model is being wired now, post-codex-preflight. **Live latent- α is the critical path and the honest open piece** — readout is proven; α (latent compile) is the hardest part, and a wrong compile means a confidently wrong organ.

4.5 The base matrix — one organ, many hosts

The pretrained organ is **base-agnostic**: Stage 1 is done once, and the woven readout (Stages 2–3) is a weights-only bolt-on, so each base is a cheap, independent paper datapoint. The set now spans **100M → 32B across six model families**, including an SSM-hybrid, to show the technique uplifts across scale **and** architecture rather than on one lucky host.

base	tier / family	why it is in the matrix
Pythia-160M / SmolLM2-135M	100M · Pythia/SmolLM	ships all training checkpoints = interp-gold; the organ-carries-the-reasoning extreme (host too small to reason alone)
OLMo-2-1B	1B · OLMo	cheap full-recipe datapoint, fully-open base; low end of the scale curve
SmolLM3-3B	3B · SmolLM	fully-open small-modern model; a clean small datapoint between 1B and 7B
OLMo-3-7B (PRIMARY)	7B · OLMo	interp anchor, full causal controls, the honest fully-open mechanism — run #1, never skipped
OLMo-3-32B	32B · OLMo	does the uplift hold / grow with base size (QLoRA @ 4-bit)
Gemma-4	mid · Gemma	portability + uplift on an already-strong, code-trained model
Qwen-3.6	mid · Qwen	portability across a second strong family
Nemotron-H-8B	8B hybrid · Nemotron	the architecture-diversity datapoint — Mamba-2 + self-attention 1. MLP hybrid; does the residual-stream coupling survive when the splice point is an SSM block, not attention? Also fully open (weights + data + recipe), so interp-friendly like OLMo

Table 1: The base matrix. 2.5 orders of magnitude (100M→32B) over six families (OLMo · Gemma · Qwen · Nemotron-H · SmolLM · Pythia), including an SSM-hybrid. Compute: woven = frozen base

1. LoRA + α + γ ; 7B/8B fit one L40S, 32B fits one A100-80 (or L40S @ 4-bit), the $\leq 3B$ tier runs anywhere. OLMo-3-7B is run #1.

4.6 What each stage proved, and the standing caveats

The evidence ladder (§4, in order): the readout works (corrupt→0, oracle de-risk); a **frozen learned organ** reads as well as the oracle; **cold co-training fails and staging fixes it**; the staged generative model uses the organ across seven rung types (corrupt→2.7% with text present); next-token CPT recruits the organ **with no reward** (architecture, not RLVR trick); and the split-brain control shows the LM wields the organ 99.9% over adversarial text (necessity, not opportunism). The standing caveats stay on the page: the organ’s soundness is **trained**, not by-construction

(the certified operators are the floor, the net is guidance — **neural proposes, certified guard narrows**); the model trusts the organ **unconditionally** (split-brain), so **α -correctness is the whole ballgame**; and the **affine wall** is real — single ternary constraints need grade-3 and XOR systems need the certified row-space organ, neither beaten by depth alone.

5 Findings — the honest arc

This is the project’s spine read in order, negatives included. The shape of the arc: the **coupling** works on a frozen LLM; RLVR and scale are **not** the levers people assume; the naive end-to-end version **fails**; a staged version **works**; and the live organ — once it became the **real** consolidated bank+composer rather than a single-faculty stopgap — **engages causally**, is **legible**, **generalizes by reduction**, and is **robust by construction**. The one thing that distinguishes all of this from an award-winning artifact that was secretly reading the labels is the causal controls, run on every result.

5.1 The checked-deductor coupling works (and what RLVR / scale actually buy)

A frozen LLM bolted to a checked deductor gains a capability it structurally lacked: **calibrated abstention**. Deterministic accuracy rises to **88.8% vs a base 1.4%** — the base essentially cannot say “I can’t conclude that.” Three sharpening results, each a “not what you’d guess”:

- **RLVR fixes calibration, not capacity.** RL adds +18–20 points of abstain-recall OOD — it tunes **when** to defer — but does not raise the underlying solvable set (arXiv:2504.13837 see §6).
- **7B $\approx$ 1B on the coupling.** Scaling the host 1B→7B does not improve the **reading** of the organ. Capacity is not the wall.
- **Grounding fixes extraction robustness, not exactness**, and the **rich-state set-valued readout makes degradation graceful and sound** — clean +58 > partial +28.7 > wrong +8 $\approx$ corrupted −3; the LM mines a partial-but-sound output and down-weights confident-wrong cells. Rich beats a collapsed point-readout by +27.5 pts on partial states.

5.2 Cold co-training fails — and staging fixes it

The decisive negative: a **learned** organ co-trained from scratch with the LM is **ignored**. The LM settles into a lazy optimum — shortcut the answer from prompt text — and the causal controls go flat. This is the same disease as SATNet’s grounding collapse (§5) and the reasoning-shortcut theorem (arXiv:2305.19951): wherever the loss admits a cheaper non-deductive optimum, it is taken. The fix is **not** to deny the LM the text; it is to **decouple** training. Two intermediate results pinned the diagnosis:

- **Oracle readout (de-risk, positive).** Feed the LM the **true** narrowed lattice through structured γ and its LM head generates the right answer **causally** — corrupt the lattice and accuracy goes to 0 **even with the full problem in the prompt**, 100% OOD. The “latch gap” was a **bandwidth** problem; structured dense γ solves it.
- **Frozen-learned-organ readout (positive, reconciling).** A **frozen, standalone-trained learned** organ reads through γ essentially as well as the oracle. So the failure was **distrust of a moving organ**, not the coupling.

5.3 The staged generative readout works (readout-isolation milestone)

Putting the recipe together first yielded a **readout-isolation** deliverable: a **learned, general** organ causally woven into generation across seven reasoning types, with the organ handed the **true** narrowed lattice. **Woven 98.8% · base 48% · text-LoRA 25%** in-distribution; OOD-phrasing 96.3%; **corrupt→2.7% with the full problem text present**. This proved the LM reads back and generates **through** a certified state it cannot get from the prompt — but it left the hard half (where does the lattice come from?) open.

5.4 ★ The consolidated woven organ engages — live, on a certified floor

The live woven model is now the **real** consolidated organ (`organ.bank_woven`): α latently compiles the host hidden into a per-cell lattice; the **composer** runs the **certified floor** (Arc/Factor/Modular/GF2/Macro)

1. the pretrained narrower + α , all **verifier-gated**; γ reads the **composed** lattice. It **engages, causally and necessarily**: true $\approx$ 100% $\rightarrow$ corrupt $\approx$ 0% (a 97–100 pt drop), lift +77 over the no-organ floor, **no-op@init** bitwise 0.0, **false-elim** 0 vs the exact verifier. The levers

are the **two-stream** asymmetry (α reads full text; the LM generates from a fact-ablated cells prompt) + **J₀ direct α -supervision** (the SATNet grounding fix). The earlier “gate stays shut, drop = 0” was a step-count under-training artifact, not a model verdict.

The structural payoff: because the certified ops run from the full domain and the meet of sound narrowings is sound, the neural/ α proposals can only sharpen **toward** the exact transformer — never below it. The injected lattice is therefore **miscompile-robust by construction**: a confidently-wrong α cannot poison it. This **subsumes** the earlier “train the organ to be robust” calibration story — robustness is now structural.

5.5 Solve-by-reduction generalizes

The cross-type reduction graph (`organ.reductions`) turns one faculty into many. Every edge is **exact-verified end-to-end against X’s own solver**; cost-routing is Dijkstra with a certified-over-approximate tie-break. On the **real** bank-woven model: a **route-required** family with **no direct faculty** (XOR-SAT) is solved 100% **via the reduction** (lift +47.5); **held-out reduction-paths** 100% (lift +37.5); base-with-LoRA-off 0% (the answer exists only through the reduced-then-solved lattice). The LM’s **recognize · route · read · decode** transfers across family and representation — the structural Karp prior, handed to the model nearly for free.

5.6 ★ The organ is legible — learned-free

A probe with **no trained decoder anywhere** (`interp_probe_bank`) reads α ’s compile by **direct set-comparison** to the exact `dedp`: overall **F1 0.90, recall 0.98** (rarely misses a true constraint; precision 0.82), per-rung `alldiff` 0.97 > coloring 0.90 > arithmetic 0.89 > equality 0.86 > ordering 0.85; candidate-set recovery 0.93–1.0 across cardinalities; the query reaches a singleton 100%. The composer’s reduced-product trace is legible by construction (which faculty fired, which neural proposals were gated). We can read **what the organ posed and what it concluded** — the interp advantage over a dense bypass is real ($n = 300$).

Scope, stated plainly. This is the **easy** regime, where the problem’s true constraint structure is handed to the composer. There the **certified floor carries answer-correctness on its own**: the probe shows answer-acc 1.0 **whether or not** α is faithful, with `corr(faithful, correct) = 0` — **no variance to correlate**. So the bottleneck test — **does α -from-hidden drive correctness?** — is **saturated / inconclusive** here. The genuinely open piece is **α -on-real-NL**: α compiling from natural language with **no provided structure**, where correctness drops and failures attribute. The `eval_real_nl` hook is wired; the result is the next frontier. **Proven: legible, engages, certified-floor-robust, reduction-generalizes. Open: α -from-real-text driving correctness.**

5.7 The coupling is residual-stream-agnostic — 7/7 bases

The graft (`organ.graft`) only ever **reads** a layer’s output hidden and **adds** to a later one through the zero-init gate, so it ports to any residual stream. Proven across 100M→32B and three architecture families, each **bitwise no-op@init** + gate-open-moves-logits:

base	architecture	layers / hidden	no-op@init
Pythia-160M	GPT-NeoX	12 / 768	✓ 0.0
OLMo-2-1B	Olmo2	16 / 2048	✓ 0.0
SmolLM3-3B	SmolLM3	36 / 2048	✓ 0.0
Gemma-4-12B	multimodal-wrapped	48 / 3840	✓ 0.0
Qwen-3.6-27B	full / linear-attn hybrid	64 / 5120	✓ 0.0
Nemotron-H-8B	Mamba-2 + attn + MLP hybrid	52 / 4096	✓ 0.0
OLMo-3-32B	Olmo3 (QLoRA 4-bit)	64 / 5120	✓ 0.0

Table 2: The 7/7 model-agnostic matrix. The Nemotron-H datapoint is decisive: the inject point is **on a Mamba-2 SSM block**, and the residual edit is still clean — every block boundary is a plain residual add, regardless of mixer.

5.8 The CPU wall, crushed (Rust hot path)

The pretrain bottleneck was the single-threaded Python exact- ded_p ground-truth path, not the GPU (the organs are under 0.5M params). `clair_fast/` (PyO3 + rayon) ports it — GIL-released in-process parallelism replaces the pickling ProcessPool — for **29–56× wall-clock** (per-call $6.7\times$ small $\rightarrow 55\times$ large), **bitwise-identical** (13,590 comparisons, 0 mismatches; a pure-Python fallback gated by `CLAIR_NO_FAST`). This unblocks the real organ-pretrain at the d^n -explosion budgets that the larger-budget organ needs.

5.9 The causal-control table — the actual contribution

Every woven result is gated behind adversarial controls. A **sound** organ guarantees a **correct** answer if the organ produced it; it does **not** prove the organ produced it. These controls do.

control	what it injects / what it proves	result
true	the real composed lattice — the intended path	pass
shuffle	a different problem’s lattice; unchanged accuracy $\rightarrow$ bypass	collapses
permute	permute candidate/value labels; tests intended semantics, not an alias	$\rightarrow 0$
corrupt	flip only the query cell’s survivors to a wrong set; must change the answer	$\rightarrow 0$
zero	no injection at all (the true zero) — the no-organ floor	lift +77

Table 3: The causal controls — exactly the ones SATNet did not run. **wired** requires corrupt/shuffle/permute to collapse **and** the gate open; **necessary** requires woven $\gg$ text-LoRA (the two-stream lever). Both must hold for **engages** — and on the consolidated organ they do.

5.10 A (explicit) vs B (latent): the bitter-lesson path that became the default

The latent path B — feed the organ **only** the LM’s hidden, no extracted factors — narrows 94% of ded_p in-distribution, is **provably equivariant** (perm/size error = 0), and after wide- N + multi-task training is sound and phrasing-robust (false-elim 8.8% $\rightarrow$ 0.9%; recall 76 $\rightarrow$ 88). It carried a 20-point OOD-by-size completeness gap vs explicit factors — but it is the path that the consolidated organ adopts and **de-risks structurally**: the certified floor now backs the latent α , so the residual α -leak can no longer cause an unsound answer, only a less-narrowed one.

5.11 Macro-deduction: log-depth on the bounded-width side

Checked **macros** give sublinear-depth sound reasoning: $\lceil \log_2 L \rceil$ macro-applications replace L base steps, **identical exact ded_p fixpoint**, **0 false-elim**, **verified to $L = 256$** . The compression is **symbolic transition-monoid composition** (graph-squaring, arXiv:2210.10749), not the net. Cost

is polynomial in $S = k^w$, so it works **only on the bounded-width side: width is the wall; bounded-width depth is a log-depth shortcut.**

5.12 Blades and the recurring null

Blades (the GA grades) help **affine generalization** specifically (grades $\rightarrow$ arities), and buy **soundness** off-distribution: zero-shot to a never-seen arity-3 relation, a generic relation-table proposer’s false-elim collapses to 0.36 while the blade’s grade-3 path stays 0.07 ($5\times$ more sound). The recurring signal across the bank: the **affine wall (3-way correlation) is the level-0 ceiling** — single ternary constraints need grade-3, XOR systems need the certified GF(2) row-space organ, and neither is beaten by depth alone. The geometric product as a generic mixer remains a clean null; its value is structural, not magical.

6 Related work and the novelty delta

GLaDOS is a **combination**, not a from-nothing primitive — every ingredient has a precedent, and being honest about which is the whole point of the honesty banner. What no one holds is the **five-way** combination, and what no one else **ran** is the discipline. We position against the four nearest neighbors.

6.1 TransNAR — the coupling cousin (but soft, and it skips the hard half)

TransNAR (arXiv:2406.09308) couples an LLM to a **frozen, pretrained** graph reasoner via zero-init gated cross-attention, trained with **pure supervised next-token, no RL**, for $>20\%$ absolute OOD gains. It validates our coupling design almost line-for-line: gate init closed, read node **and** edge structure by concat+linear (pooling loses), reasoner frozen during coupling, and — a fix it handed us — **randomized positional encoding is load-bearing**: without it the hybrid is thresholded by the base LM’s OOD score, a candidate cause of our cold-weave shortcut. Two deltas: TransNAR’s reasoner is **soft** (a CLRS-pretrained net, wrong OOD — its index/search failures are exactly that), where ours is **output-checked $\rightarrow$ sound**; and its reasoner ingests a clean external graph the LLM never produces. Our α (hidden $\rightarrow$ typed program) is precisely the input-production step TransNAR sidesteps — the unprecedented half.

6.2 Coconut — our mechanism, emergent and unchecked

Coconut (arXiv:2412.06769) makes a **latent** “continuous thought” loop load-bearing under ordinary LM loss — **but only with a multi-stage curriculum** that progressively replaces text CoT with latent thoughts. **Remove the curriculum and the module is ignored** (GSM8K 34.1% $\rightarrow$ 14.4%; ProntoQA 99.8% $\rightarrow$ 52.4%). This is the cleanest external analogue of our cold-co-train failure **and** of our staging fix: LM loss **can** exploit a reasoning module, but a naive single-stage objective collapses to the shortcut. The delta: Coconut’s latent loop is a **free-form superposition-elimination** with no check — exactly our mechanism, but **emergent and unverified**. Ours is checked, so its degradation is **abstention**, not silent error.

6.3 SATNet — the cautionary tale, and the discipline lesson

SATNet (arXiv:1905.12149) was a differentiable MAXSAT layer that appeared to learn logic end-to-end, including “visual Sudoku” with no digit labels. The grounding critique (arXiv:2312.11522) showed the “learned logic” was **label leakage**: the loss term on input bits backpropagated the true digit labels straight into the CNN through an unmasked side channel. With that channel masked, visual Sudoku collapses to **0.0% $\pm$ 0.0%** over 10 seeds; a plain FC net **beat** the SATNet model (72.1%). The reasoning-shortcut theorem (arXiv:2305.19951) then proved the general result: **reasoning shortcuts are optima of the loss** — a NeSy model can attain perfect accuracy with unintended concept semantics, right for the wrong reasons.

The discipline lesson. SATNet is the field’s most reproducible result about systems like ours, and it is a **negative**: a **sound** organ can still be **bypassed**, and end-to-end differentiability **hides** it. The soundness check is therefore **necessary but not sufficient** — it buys correctness-given-the-organ, never load-bearingness. Our edge over the whole field is not the organ; it is the **causal controls (corrupt / shuffle / permute)** — exactly the controls SATNet did not run — gating every claim, plus isolating α with its own grounding test before co-training, and reporting over seeds. The dense- γ oracle result is the SATNet test **passed**.

6.4 RLTT / LSRL — process reward, and the grader they wanted is our organ

The process-reward line fixes a credit-assignment bug we will hit: outcome-only GRPO credits **only the terminal state** and so **cannot reward an organ’s internal steps** (RLTT, [arXiv:2602.10520](#) LSRL, Findings EMNLP 2025). RLTT broadcasts one outcome reward across loop steps parameter-free; LSRL builds a genuine per-step reward by decoding and grading each latent state — **with a hackable GPT-nano judge**, and it explicitly asks future work for “a symbol-aware PRM verifying each step.” **That is the GLaDOS organ.** Process-reward = candidate-set cardinality drop; step-quality = soundness of the narrowing. Exact, unhackable, free. The novelty delta in one line: **TransNAR is soft, RLTT has no verifier, LSRL’s judge is hackable — ours is exact.**

6.5 The five-way combination nobody holds

Surveying the differentiable-symbolic-module cell (NeSy $\times$ LLM survey [arXiv:2508.13678](#) nearest named cousins DiLA and the VSA/HRR readout [arXiv:2502.01657](#) reward-side RLSF [arXiv:2405.16661](#)), no single system holds all five of:

property	who is missing it
(i) woven in the forward pass	DiLA, RLSF, tool-LLMs are external
(ii) differentiable, dense structured readback	tool / abductive / post-hoc-verifier methods are not
(iii) output-checked $\rightarrow$ sound	SATNet, TransNAR, AutoCoNN, the VSA readout are soft
(iv) a per-problem compiled typed program	DeepProbLog/LTN/SATNet use fixed/learned-but-fixed rules
(v) verifier / RLVR-trained	only RLSF + the tool-verifier line — and those are not woven

Table 4: The honest novelty claim: **first to weave a checked, per-problem-compiled deductor into a pretrained LLM and verifier-train it.** The novelty is narrow and integrative — a combination plus the soundness twist, not a new primitive. The defensible edge is the discipline.

7 Toward real training, and the decisive open experiment

The toy is built and it works — but it is profoundly **undertrained** (1B host, synthetic CSPs). The honest next step is real RLVR on standardized verifiable reasoning, with the organ as a **tested** component, on a base model where a gain **means** something.

7.1 The RLVR pivot

Two literature findings reshape **how** we test the organ before **whether** it helps:

- **Use OLMo as the base.** “Spurious Rewards” ([arXiv:2506.10947](#)) shows random/wrong rewards recover most RLVR gain on Qwen-Math but **not** on Llama/OLMo — Qwen’s RLVR partly measures pretraining. On OLMo a gain is real signal. We standardize on **OLMo-3-Base-7B**

(Base, not Instruct: RL headroom + an honest control; our 7B $\approx$ 1B coupling finding means 7B buys LM competence, not better reading). A random-reward and incorrect-label **dummy arm** is mandatory — on OLMo it should do nothing; if the organ-augmented model suddenly gains under random reward, that exposes an amplifiable shortcut.

- **Report pass@k, not just pass@1, with a ProRL-style control.** RLVR mostly **elicits** rather than **expands** (arXiv:2504.13837): it raises pass@1 and can **lower** pass@k. The capability question is whether the organ behaves like **distillation** (lifts pass@k, expands the boundary) or like plain RLVR (sharpens the prior). A prolonged, KL-controlled, task-diverse RLVR arm (ProRL) is needed so we do not credit the organ for beating a deliberately weak baseline.

The substrate is de-risked: the base TRL-GRPO loop (`rlvr_pipeline.py`) **runs and learns** on one L40S (reward EMA 0.20 $\rightarrow$ 0.72 over 250 steps on `chain_sum`), with `use_vllm=False` for an organ-compatible dynamic-depth rollout, Dr.GRPO, $\beta = 0$, LoRA all-linear. The organ slots in by swapping the policy model; reward, dataset, and config are unchanged. Fuel is reasoning-gym (arXiv:2505.24760) plus ZebraLogic (arXiv:2502.01100) / SATBench for CSP transfer and a MATH slice for cross-domain. The premise is vindicated by Logic-RL (arXiv:2502.14768): a 7B trained **only** on procedural Knights-and-Knives transferred to AIME/MATH.

7.2 The organ as process reward

Because outcome-only GRPO **falsely nulls** a looped organ (it never credits internal steps), a process/trajectory-reward arm is not optional — it is what makes a GRPO null **interpretable**. And the organ **is** the grader: per-step progress = candidate-set cardinality drop, per-step quality = soundness of the narrowing. We adopt both published arms (RLTT broadcast, parameter-free; LSRL exact-graded), honoring the depth floor (process supervision is null below $r = 8$ recurrent steps) and keeping the outcome reward primary (0.7 outcome / 0.3 process) so the per-step signal shapes rather than dominates.

7.3 The decisive open experiment: architecture or trick?

The single most decisive experiment. The organ supplies **correctness**; next-token CPT optimizes **likelihood**. On generic prose the two diverge, so the gradient that would route generation through the organ may be absent (the cold-weave failure). **Does ordinary next-token continued-pretraining exploit the organ — making it a general architecture — or does only targeted RL open the gate — making it a trick?** No published work tests next-token CPT on a **checked deductor**.

The design is a **2 $\times$ 3 grid**: {next-token CPT, process-RLVR} $\times$ {reasoning-dense + non-shortcuttable, reasoning-dense + shortcuttable, general prose}. Starting from a checkpoint M $\star$ where the organ is already causally load-bearing (staged recipe), continue-train with **ordinary next-token loss only** and track the **exploitation gap** $G(t) = \text{loss}_{\text{off}} - \text{loss}_{\text{on}}$ — next-token loss with the organ zeroed vs on, over CPT steps. **Exploited** = $G(t)$ grows monotonically and concentrates on deduction-determined (low-entropy, verifier-checkable) tokens, the γ gate opens from 0, the corrupt $\rightarrow$ 0 penalty **rises**, and OOD verifiable accuracy improves **with no verifier reward in the loop**. **Null (RL-only trick)** = $G(t)$ flat/shrinking and the gate stays shut while the process-RLVR arm on the same data **does** open it. The literature is genuinely split here: the get-go position (arXiv:2502.19402, which proposes our exact reasoner+memory architecture and bets on RL) and SFT-memorizes/RL-generalizes (arXiv:2501.17161) predict the trick; procedural-knowledge (arXiv:2411.12580), front-loading (arXiv:2510.03264), and distillation (arXiv:2504.13837 Fig. 7) predict the architecture — **given the right (reasoning-dense, non-shortcuttable) data**. The 2 $\times$ 3 grid adjudicates it on our own organ.

7.4 Future architectures

Two host shapes marry the organ better than a dense autoregressive stack:

- **DiffusionGemma — organ $\leftrightarrow$ denoising.** A diffusion/denoising LM already runs an **iterative refinement** over the whole sequence; the organ’s monotone narrowing is a natural per-step constraint **projection** inside the denoising loop (narrow the candidate set, then denoise toward it). The organ’s recurrence and the model’s iteration become **one** loop rather than two nested ones — and the monotone exit signal gives the denoiser a principled, sound stopping criterion.
- **MoE — organ as a routed expert.** In a mixture-of-experts host, the organ is one expert the router can dispatch to on reasoning-dense tokens. This makes “use the organ” a **learned routing decision** trained by the same load-balancing machinery, and it bounds the organ’s cost to the tokens that need it — a clean answer to “when should the model reason through the organ” that the router learns rather than the gate.

The consolidation endpoint remains the same in all cases: re-run OLMo’s own open recipe (Dolma continued-pretrain + instruction-tune + RL-Zero) **with the organ in place**, so organ-use folds into general behavior rather than staying a bolted-on skill. Whether **ordinary** training does that is the architecture-vs-trick question above — and it is the experiment that decides what GLaDOS actually is.

Lineage / references (arXiv ids inline above). LDT 2605.08605 · CliffordNet 2601.06793 · automata-shortcuts 2210.10749 · TransNAR 2406.09308 · Coconut 2412.06769 · SATNet 1905.12149 · SATNet grounding critique 2312.11522 · reasoning shortcuts 2305.19951 · RLTT 2602.10520 · LSRL (Findings EMNLP 2025) · looped transformers 2502.17416 · two-scale recurrence 2509.23314 · RLVR-elicits 2504.13837 · spurious rewards 2506.10947 · get-go RPT 2502.19402 · SFT-memorizes-RL-generalizes 2501.17161 · emergent-under-CPT 2506.00288 · procedural-knowledge 2411.12580 · front-loading reasoning 2510.03264 · Logic-RL 2502.14768 · reasoning-gym 2505.24760 · ZebraLogic 2502.01100 · DiLA (OpenReview uh3ZO2izyr) · VSA/HRR readout 2502.01657 · RLSF 2405.16661 · NeSy×LLM survey 2508.13678. Full notes + URLs in [clairnets/notes/](#).